CS 766 Midterm Progress Report

Automatic Photo Augmentation from a Single Image

Team Members: Jiapeng Zeng, Heyan Zhang, Lanxi Zhang

1 Introduction

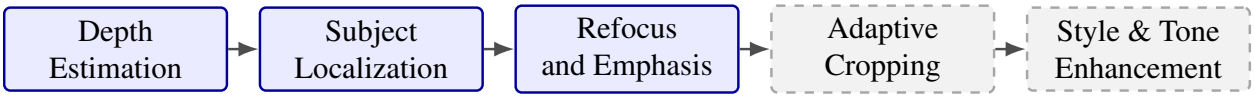
The goal of our project is to build a system that can automatically improve a photo after it is taken, without requiring manual editing from the user. Given a single input image, our system tries to do three things: emphasize the subject using depth-aware synthetic defocus, improve composition through adaptive cropping, and apply a light style or tone enhancement at the end.

The overall idea from our original proposal has not changed. We still want to treat these three steps as one unified pipeline rather than three unrelated tools. The main difference at this point is that we have moved from the planning stage to an actual working implementation.

2 Current Progress

2.1 Pipeline Overview

Our overall goal is to build a full automatic photo augmentation pipeline. The complete pipeline is designed to include the following five stages:



1. **Depth estimation.** We use Depth-Anything V2 to estimate a dense monocular depth map from a single input image, with a lightweight pseudo-depth fallback when the learned model is unavailable.
2. **Subject localization.** We compute a coarse saliency map by combining depth prior, local contrast, edge strength, saturation, and center bias. In auto mode, this is further combined with HOG-based people detection to decide whether to emphasize people or the overall scene.
3. **Subject emphasis and refocus.** Using the subject mask and the estimated depth map, we apply depth-aware background blur while preserving the subject region as sharp as possible.
4. **Adaptive cropping.** A rule-based module generates a crop box based on subject position, subject size, and target aspect ratio, with a gentle rule-of-thirds bias to improve composition while keeping enough context.
5. **Style and tone enhancement.** A lightweight filter module applies tone-curve adjustment, color grading, saturation tuning, film grain, and vignette effects to produce the final stylized output.

At the current midterm stage, our main progress has been on the first three stages of the pipeline. We are currently developing the adaptive cropping stage, and the remaining work in the second half of the semester will focus on completing the cropping and style enhancement stages and integrating all five stages into a more stable end-to-end system.

2.2 Implementation Status

At this stage, we have implemented the first three main components of the pipeline and obtained preliminary results for depth estimation, subject localization, and refocusing. The later stages, especially adaptive cropping and style enhancement, are still being refined and will be completed in the second half of the semester.

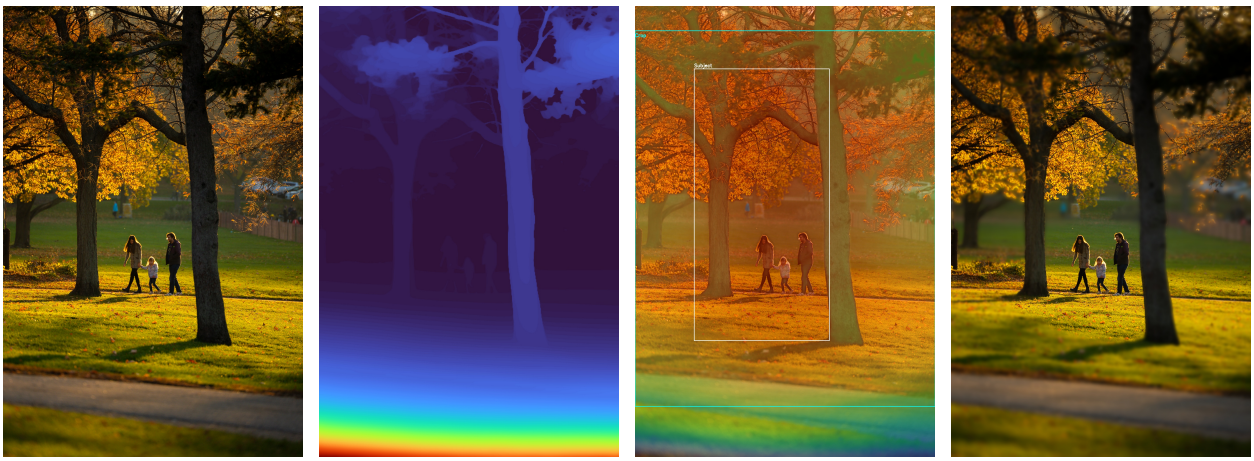
We have also begun organizing materials for a simple project webpage that will present the pipeline and selected example results more clearly.

3 Results So Far

So far, the main result is that we have obtained some preliminary outputs from the first three stages of the pipeline: depth estimation, subject localization, and refocusing.

On simpler examples, especially images with one relatively clear subject, the current system can already produce usable depth maps and reasonable subject-prior visualizations. Using these intermediate results, our refocusing stage is able to blur the background while keeping the subject relatively clear in some cases.

These results are still preliminary, and the later stages such as adaptive cropping and style enhancement are still under development. However, the current intermediate outputs already show that the first part of the pipeline is moving in the right direction.



(a) Input (b) Depth (c) Subject (d) Refocus

Figure 1: Preliminary outputs from the first three stages of our pipeline: input image, estimated depth, subject localization, and refocused output.

4 Challenges Encountered

4.1 Depth Estimation at Object Boundaries

One of the most obvious problems in our current results appears at object edges. The estimated depth map is often smooth near foreground or background boundaries and does not always preserve sharp transitions. When we use this depth map to control the blur radius, the defocus effect can look unnatural around object contours, with a visible halo or seam where the blur changes too abruptly. This is especially noticeable for hair, clothing edges, or thin objects in front of a blurred background. As a result, the output can look more synthetic than realistic, especially under large-aperture settings.

4.2 Cropping: Visually Awkward and Inconsistent Results

Another issue is that our current cropping results are still not very satisfying. In many images, it is not always clear whether the system should emphasize a person or the overall scene. Because of this, the crop can feel

inconsistent in both position and size. Our current cropping heuristic has a rule-of-thirds bias, but following this kind of composition rule in a mechanical way does not always produce a crop that actually looks natural. In some cases, the crop is too weak and barely changes the original image, while in other cases it is too aggressive and removes context that should have been kept.

4.3 Sensitivity to Image Type

Another difficulty is that the current system does not behave equally well across different kinds of input images. It works more reliably on images with one relatively clear foreground subject, but becomes less stable on scenes with multiple possible subjects, cluttered backgrounds, or weak visual contrast. This makes it hard to write one set of rules that works well on all kinds of images.

4.4 Difficulty of Evaluation

It is also hard to evaluate our results objectively, since photo quality is largely subjective. There is no single metric that can capture whether an output looks better overall, because an image might improve in one aspect but look worse in another. Different people may also have different opinions on what looks good.

4.5 Error Propagation

Because the pipeline is sequential, mistakes in early stages carry forward. An inaccurate depth map makes the refocusing stage look wrong, and an unreliable subject estimate can also affect later stages such as cropping and enhancement. Each module can work reasonably well on its own, but getting the whole pipeline to behave consistently across many different types of images is still the main challenge.

5 Changes from the Original Proposal

The overall goal of the project is still the same. We still want an automatic pipeline that improves focus, composition, and style from a single input image. However, a few things have changed in how we are approaching the project.

- **Using pre-trained components.** We are using Depth-Anything V2 for depth estimation, together with a heuristic multi-cue method for subject localization, rather than training new models from scratch. This keeps the project manageable while still allowing us to focus on pipeline design, integration, and evaluation.
- **A simpler evaluation plan.** The proposal mentioned a user study as one possible evaluation direction. At this point, we expect to focus more on qualitative comparisons across a diverse set of test images instead of a formal user study. We may still collect some informal feedback, but the main goal is to show clear before/after comparisons across different image categories and to present representative results through our webpage.

6 Next Steps

For the rest of the semester, we plan to:

- **Complete the adaptive cropping stage.** Refine the crop so that it is more sensitive to image content, especially the difference between person-focused and scene-focused images, and add better control over crop scale.
- **Complete the style and tone enhancement stage.** Add a lightweight enhancement module for tone adjustment, color grading, and other basic stylistic effects so that the final output looks more polished.
- **Complete end-to-end integration.** Make the full pipeline more stable by connecting all five stages more reliably and reducing failure cases caused by error propagation across modules.
- **Test on more images.** Run the current system on a wider range of images, including portraits, landscapes, product photos, and multi-subject scenes, and document both successful cases and failure cases.
- **Prepare the webpage and final report.** Organize clearer pipeline diagrams, example outputs, and a discussion of our design choices, limitations, and lessons learned.